

## 12. class, Electrodynamics and Optics

### Page 892, Task 10.

Verify by substitution that the following equations are solutions to Equations 33.19 and 33.20, respectively:

$$E = E_{\max} \cos(kx - \omega t)$$

$$B = B_{\max} \cos(kx - \omega t)$$

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} \quad (33.19)$$

$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2} \quad (33.20)$$

### Page 892, Task 18.

Assuming the antenna of a 10.0-kW radio station radiates spherical electromagnetic waves, (a) compute the maximum value of the magnetic field 5.00 km from the antenna and

(b) state how this value compares with the surface magnetic field of the Earth.

### Page 893, Task 20.

At one location on the Earth, the rms value of the magnetic field caused by solar radiation is  $1.80 \mu\text{T}$ . From this value, calculate

(a) the rms electric field due to solar radiation,

(b) the average energy density of the solar component of electromagnetic radiation at this location, and

(c) the average magnitude of the Poynting vector for the Sun's radiation.

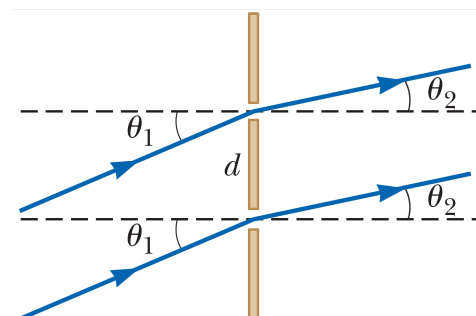
### Page 893, Task 21.

A 25.0-mW laser beam of diameter 2.00 mm is reflected at normal incidence by a perfectly reflecting mirror. Calculate the radiation pressure on the mirror.

### Page 978, Task 9.

Coherent light rays of wavelength  $\lambda$  strike a pair of slits separated by distance  $d$  at an angle  $\theta_1$  with respect to the normal to the plane containing the slits as shown in the figure. The rays leaving the slits make an angle  $\theta_2$  with respect to the normal, and an interference maximum is formed by those rays on a screen that is a great distance from the slits. Show that the angle  $\theta_2$  is given by (where  $m$  is an integer.)

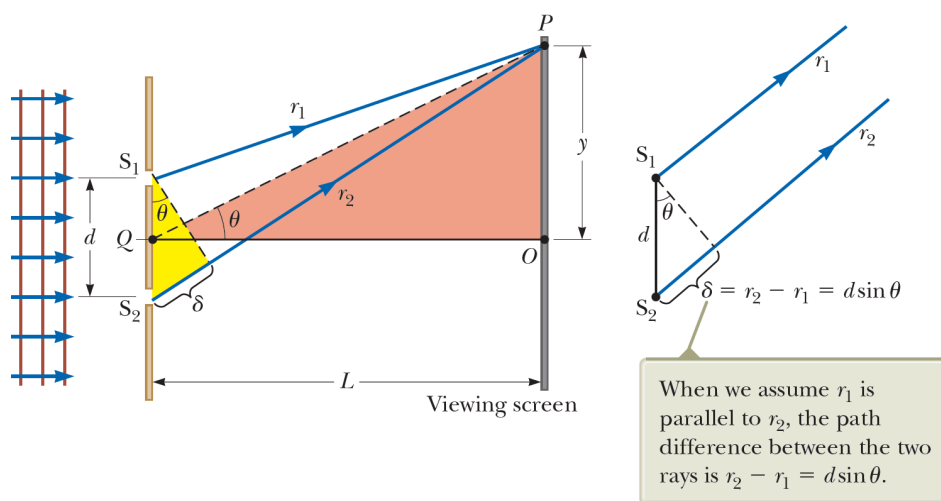
$$\theta_2 = \sin^{-1} \left( \sin \theta_1 - \frac{m\lambda}{d} \right)$$



**Page 979, Task 16.**

Show that the distribution of intensity in a double-slit pattern is given by

$$I = I_{\max} \cos^2 \left( \frac{\pi d \sin \theta}{\lambda} \right)$$



Begin by assuming that the total magnitude of the electric field at point P on the screen in the figure is the superposition of two waves, with electric field magnitudes

$$E_1 = E_0 \sin \omega t \quad E_2 = E_0 \sin (\omega t + \phi)$$

The phase angle  $\phi$  in  $E_2$  is due to the extra path length traveled by the lower beam in the figure. Recall from Equation 33.27 that the intensity of light is proportional to the square of the amplitude of the electric field. In addition, the apparent intensity of the pattern is the time-averaged intensity of the electromagnetic wave. You will need to evaluate the integral of the square of the sine function over one period. Refer to Figure 32.5 for an easy way to perform this evaluation. You will also need the trigonometric identity

$$\sin A + \sin B = 2 \sin \left( \frac{A + B}{2} \right) \cos \left( \frac{A - B}{2} \right)$$

**Page 979, Task 20.**

A soap bubble ( $n = 1.33$ ) floating in air has the shape of a spherical shell with a wall thickness of 120 nm.

- What is the wavelength of the visible light that is most strongly reflected?
- Explain how a bubble of different thickness could also strongly reflect light of this same wavelength.
- Find the two smallest film thicknesses larger than 120 nm that can produce strongly reflected light of the same wavelength.